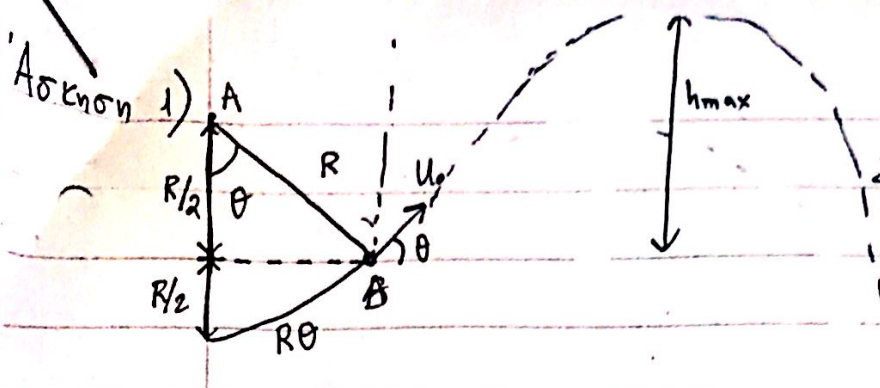
Λύσεις Εξέτασης 29/10/21 ①

Άσκηση 1)



$\Sigma F_{rad} = m \frac{U_{min}^2}{R} \Rightarrow mg = m\frac{U_{min}^2}{R}$

A) $U_{min} = \sqrt{gR}$ (2,5μ)

B) α) Η γωνία θ μπορεί να υπολογιστεί από $\cos\theta = \frac{R/2}{R} = \frac{1}{2}$

$\Rightarrow \theta = 60°$ (4,5μ)

β) Η ταχύτητα στο σημείο Β είναι διπλάσια της U_{min} ⇒

$U_0 = 2\,U_{min} = 2\sqrt{gR}$.

Γ) γ) $h_{max} = \frac{U_0^2 \sin^2\theta}{2g} = \frac{4gR}{2g}\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3R}{2}$ (2μ)

δ) $U_B = \sqrt{U_0^2 + 2gR/2} = \sqrt{4gR + gR} = \sqrt{5gR}$ (2μ)

ε) Βεληνειές $= \frac{U_0\cos\theta}{g}\left(U_0\sin\theta + \sqrt{U_0^2\sin^2\theta + gR}\right)$

$= \frac{2\sqrt{gR}\cdot 1/2}{g}\left(2\sqrt{gR}\frac{\sqrt{3}}{2} + \sqrt{4gR\left(\frac{3}{4}+\frac{1}{4}\right)}\right)$

$= \frac{\sqrt{gR}}{g}\left(\sqrt{3gR} + \sqrt{4gR}\right) = R(\sqrt{3}+2) = 3{,}73\,R$

περίπου $4R$. (3,5μ)

Δ) $t_{ολικό} = \frac{U_0\sin\theta + \sqrt{U_0^2\sin^2\theta + gR}}{g} = \frac{2\sqrt{gR}\frac{\sqrt{3}}{2} + \sqrt{4gR\left(\frac{3}{4}\right)} + gR}{g}$

$x = \frac{1}{2}a t^2 \Rightarrow 4R = \frac{1}{2}a\, t_{ολικό}^2$

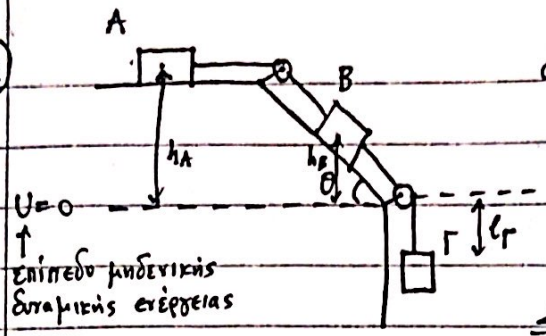
$\Rightarrow a = \frac{8R}{t_{ολικό}^2}$

$\boxed{t_{ολικό} = \sqrt{\frac{R}{g}}(\sqrt{3}+2)}$

(7,5μ)

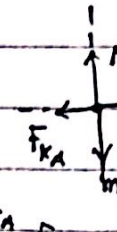
$\Rightarrow a = \frac{8R}{\frac{R}{g}(\sqrt{3}+2)^2} = \frac{8}{(\sqrt{3}+2)^2}\,g$

Σκηνή 2)



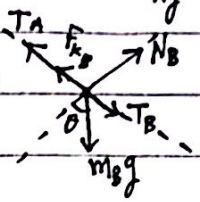
α' Τρόπος)

Σώμα Α:



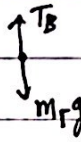
$$T_A - \mu_K m_A g = m_A a \quad (1)$$

Σώμα Β:



$$T_B + m_B g \sin\theta - \mu_K m_B g \cos\theta - T_A = m_B a \quad (2)$$

Σώμα Γ:



$$m_\Gamma g - T_B = m_\Gamma a \quad (3)$$

$$\Rightarrow \boxed{a = \frac{m_\Gamma + m_B(\sin\theta - \mu_K\cos\theta) - \mu_K m_A}{m_A + m_B + m_\Gamma} g} \quad (4)$$

$$\Rightarrow \boxed{u = \sqrt{2gd\,\frac{m_\Gamma + m_B(\sin\theta - \mu_K\cos\theta) - \mu_K m_A}{m_A + m_B + m_\Gamma}}} \quad (5)$$

β' Τρόπος) $W_{tot} = \Delta K$, $W_{tot} = (T_A - F_{K_A})d + (T_B - F_{K_B} + m_B g\sin\theta - T_A)d + (m_\Gamma g - T_B)d$

$$W_{tot} = -\mu_K m_A g d - \mu_K m_B g\cos\theta d + m_B g\sin\theta\, d + m_\Gamma g d$$

$$\Delta K = \frac{1}{2}(m_A + m_B + m_\Gamma)u^2 \Rightarrow \boxed{u = \sqrt{2gd\,\frac{m_\Gamma + m_B(\sin\theta - \mu_K\cos\theta) - \mu_K m_A}{m_A + m_B + m_\Gamma}}} \quad (6)$$

γ' Τρόπος) $E_1 + W_{other} = E_2 \Rightarrow K_1 + U_1 + W_{other} = K_2 + U_2$

$$0 + 0 + 0 + m_A g h_A + m_B g h_B - m_\Gamma g \ell_\Gamma - F_{K_A} d - F_{K_B} d = \frac{1}{2}(m_A + m_B + m_\Gamma)u^2 + m_A g h_A + m_B g(\ell_B - d\sin\theta) - m_B g(\ell_B + d)$$

$$\Rightarrow -\mu_K m_A g d - \mu_K m_B g\cos\theta\, d = \frac{1}{2}(m_A + m_B + m_\Gamma)u^2 - m_B g d\sin\theta - m_\Gamma g d$$

$$\Rightarrow \boxed{u = \sqrt{2gd\left(\frac{m_\Gamma + m_B(\sin\theta - \mu_K\cos\theta) - \mu_K m_A}{m_A + m_B + m_\Gamma}\right)}} \quad (7)$$

Α) Εάν Γ έχει σταθερή ταχύτητα $a = 0$ $\Rightarrow$ Από σχέση (4):

$m_\Gamma = \mu_K m_A - m_B(\sin\theta - \mu_K\cos\theta)$ (2,5μ) και σχέση (1) $T_A = \mu_K m_A g$. (2,5μ)

Βάρος: $W_\Gamma = m_\Gamma g$.

Β) Από σχέση (4): $(m_A + m_B + m_\Gamma)a = m_\Gamma g + m_B g(\sin\theta - \mu_K\cos\theta) - \mu_K m_A g$

$m_\Gamma(a - g) = m_B g(\sin\theta - \mu_K\cos\theta) - m_B a - \mu_K m_A g - m_A a$

$\Rightarrow m_\Gamma = \left[m_B g(\sin\theta - \mu_K\cos\theta) - m_B a - \mu_K m_A g - m_A a\right]/(a - g)$ (2,5μ)

Γ) Βλέπε σχέσεις (5), (6) και (7) → (10μ) ←

Σχέση (1) και (3)

$T_A = m_A a + \mu_K m_A g$

$T_B = -m_\Gamma a + m_\Gamma g$. (2,5μ)